SOIL EROSION AND CONSERVATION · NOTE CM02

Conservation Farming — Contour Farming, Strip Cropping and Conservation Tillage

Course topics: CAE1.3 L9, SWCE4.5 L8, SWCE7.4 L10

Contour farming, strip cropping and conservation tillage conserve soil and water and reduce erosion on farmland; experiments show they cut both soil and water losses.

Contour farming

- All field operations — ploughing, planting, cultivating — run **across** the slope, not up and down it. The small ridges and plant stems in contoured rows hold water and stop runoff and erosion.
- Most effective with row crops, also useful for small grains — the simplest and easiest mechanical conservation measure.
- **Guide lines:** on land without terraces or bunds, first lay level contour guide lines across the hillside with a levelling instrument. On soils that absorb water slowly, give the line a 0.1–0.2% slope toward the waterway instead of making it dead level. Till parallel to the lines; one line serves a small field, large fields need more.
- **Limitations:** on steeper slopes or in intense rain, rows breach and gullying increases, worse as water accumulates downslope. Surface sealing, lower infiltration and loss of depression storage as the soil settles reduce effectiveness.
- **Effect:** with grassed waterways, contour cultivation cut watershed runoff by 75–80% early in the season and 20% by the end of the year — about 66% on average annually. On steeper slopes add strip cropping, terracing or bunding.

Strip cropping

Alternate strips of close-growing and intertilled crops in the same field — more intensive than contouring, less than terracing or bunding, and often used with them. It is a combination of crop rotation, contour farming, cover cropping, conservation tillage and stubble mulching. With contour tillage or terracing it breaks the slope length, checks runoff velocity, filters soil out of runoff, helps rain soak in, and protects the bare intertilled strips.

Type	Layout and use
Contour strip cropping	Strips on the contour, at right angles to the natural slope, cropped in a definite rotation
Field strip cropping	Uniform-width strips perpendicular to the erosive force — across the general slope (not necessarily on contour) for water erosion, where topography is too irregular for contour strips; parallel strips across the prevailing wind for wind erosion
Buffer strip cropping	Grass or legume strips between contour strips of crops in regular rotation, on irregular slopes; wide or narrow, even or variable width, placed on steep or badly eroded parts — more protection than solid planting

- **Rotation:** close-growing perennial grasses and legumes alternating with grain and intertilled crops. In dry-farming areas, a sequence of row and small-grain crops, with fallow for efficient moisture use

and enough residue. Leave all natural waterways as grassed waterways.

- **Layout:** lay guide lines as for contouring, or use a terrace or bund ridge as the guide. Strips must be wide enough for convenient farming but not so wide that soil is lost — depending on slope, rotation and erodibility; on terraced or bunded land, match the terrace spacing.
- **Methods:** both edges on the contour; one or more uniform-width strips from a key contour line; or alternate uniform and variable (buffer-correction) strips. Permissible deviation: 30–60 cm above or below the true contour in any 30 m.

Land slope (%)	Good soil (m)	Fair soil (m)	Poor soil (m)
1–3	51	42	33
4–6	42	33	25
7–9	33	25	17
10–22	25	17	17

Table 14.5 — maximum strip widths.

Conservation tillage

- Ploughing, disking, harrowing and cultivating affect erosion through aggregation, surface sealing, infiltration and resistance to wind; excessive or wrong tillage makes soil more erodible.
- **Mulch tillage (stubble mulching):** leaves much of the previous crop's residue on or near the surface. Three benefits: the mulch breaks the fall of raindrops and reduces dispersion; leaves, stems and roots impede flow and control sheet erosion; and the open soil structure promotes infiltration.
- In arid and semi-arid regions, prepare the seedbed with subsurface tillers — V-shaped sweeps, straight-blade tillers and rod weeders. In heavy mulch, cut or turn the residue with a disc plough or one-way disc harrow.
- **Listing and ridge planting:** in low-rainfall areas with short intense storms on gently sloping fields, contouring alone may do, using listers (double mouldboard ploughs). A device that builds earth dikes at intervals in the furrow — **basin listing** — controls drainage and raises the furrow's conservation value.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. On slowly permeable soils, contour guide lines are given a slope toward the waterway of

(a) 0%
(b) 0.1–0.2%
(c) 1–2%
(d) 5%

Q2. Contour cultivation with grassed waterways reduced annual runoff on average by about

(a) 20%
(b) 40%
(c) 66%
(d) 90%

Q3. Maximum strip width on 1–3% slope with good soil (Table 14.5) is

(a) 25 m
(b) 33 m
(c) 42 m
(d) 51 m

Q4. Strip cropping where topography is too irregular for contour strips is

(a) buffer (b) field
(c) contour (d) wind

Q5. Permissible deviation of a strip from the true contour is

(a) 5–10 cm in 30 m (b) 30–60 cm in 30 m
(c) 1 m in 10 m (d) 2 m in 30 m

Q6. A lister is a

(a) disc harrow (b) double mouldboard plough
(c) sweep (d) rod weeder

Q7. Making small dikes at intervals in lister furrows is called

(a) contour ridging (b) basin listing
(c) tie ridging (d) bedding

Answers

Q1 — (b) 0.1–0.2%
Q2 — (c) 66%
Q3 — (d) 51 m
Q4 — (b) field
Q5 — (b) 30–60 cm in 30 m
Q6 — (b) double mouldboard plough
Q7 — (b) basin listing